Characteristics of MOSFET

Aim of the experiment:

- To study the transfer characteristics of MOSFET and find out the threshold voltage.

- To study the Drain characteristics of MOSFET.

THEORY

Transistor is a 3-terminal and 2-junction (p-n) device. Its name comes from transfer + resistance. The current through 2 terminals can be controlled by I/V through the other 2 terminals.

Broad classification –

a) BJT (Bipolar Junction Transistor) – Current-controlled device

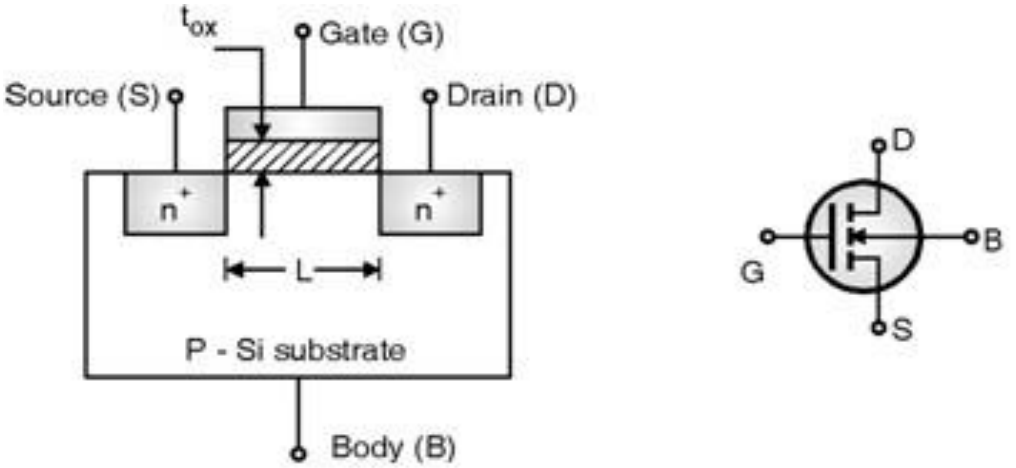
b) FET (Field Effect Transistor) – Voltage-controlled device

   i) JFET (Junction FET) – n-channel is already formed

   ii) E-MOSFET (Metal Oxide Semiconductor FET) – n-channel has to be formed

A FET has three terminals as:

1) Source (S): The terminal through which the majority charge carriers enter the channel. Conventionally, current entering at S is designated by I_S.
2) Drain (D): The terminal through which the majority charge carriers leave the channel. Conventionally, current entering the channel at D is designated by I_D. Drain-to-source voltage is V_{DS}.
3) Gate (G): the terminal that modulates the channel conductivity.



t_{ox} is the thickness of the oxide layer (generally SiO_2), L is the channel length

Figure 1. Basic structure of MOSFET

MOSFET is a field effect transistor whose drain current (I_D) is controlled by the voltage on the gate (see Figure 1). MOSFET has a much higher input impedance because of the very smallgate leakage current. It has wide applications in the field of modern-era digital circuits. It is used in the amplification of the signal and switching action. Its major advantage over BJT in switching is that it is much more power efficient at high-frequency switching. It has a +ve temperature coefficient, so it is thermally stable.

MOSFET is of 2 types: a) enhancement type - n & p type

b) depletion type - n & p type

The experiment will be done with n-enhancement type MOSFET.

Depletion/Space Charge Region –

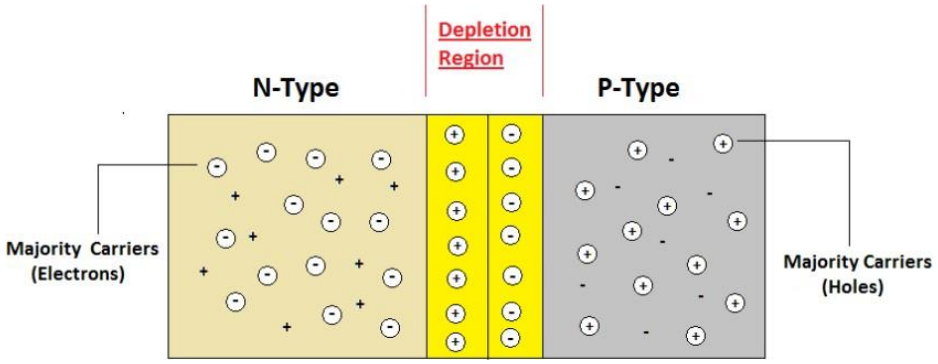


Figure 2. Depletion region in a p-n junction

There is a greater concentration of holes in the P-region and electrons in the N-region. This difference in concentration establishes a mismatch across the junction resulting in majority carrier diffusion. Holes diffuse from P to N and electrons from N to P regions causing the recombination. This recombination of electrons and holes produces a narrow region at the junction called the depletion region as shown in Figure 2.

N-channel formation and Pinch-off phenomena

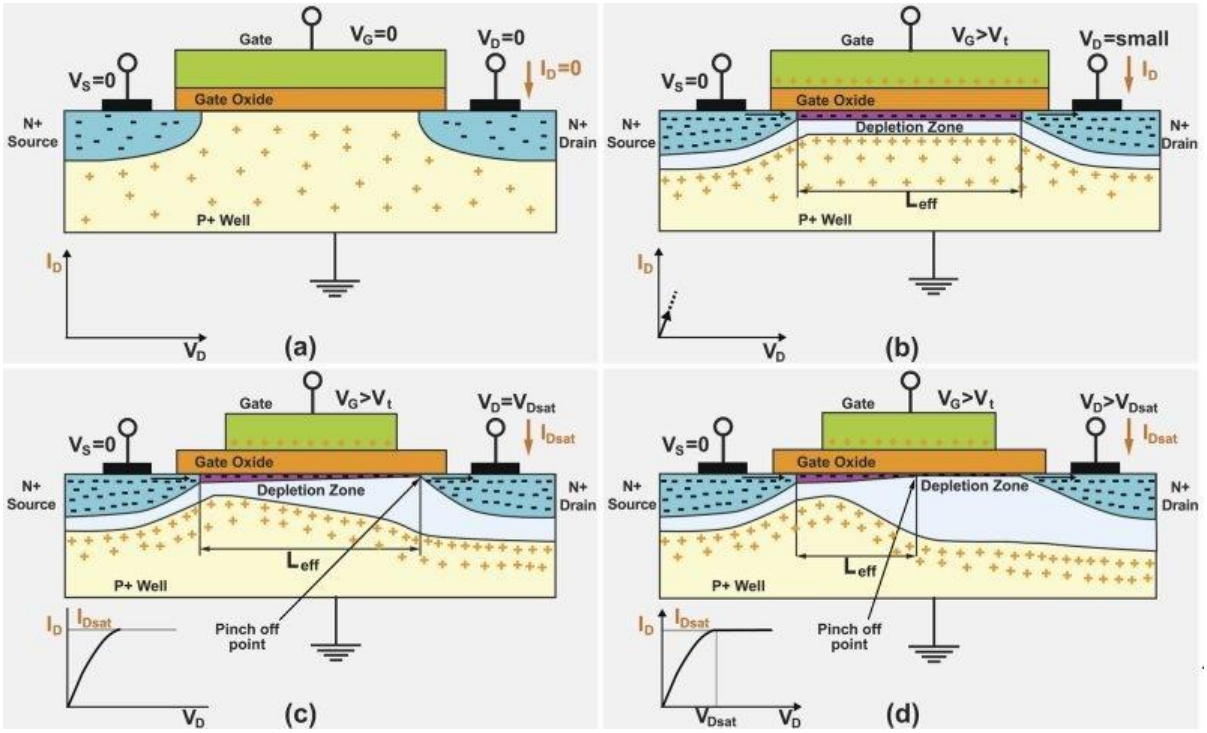
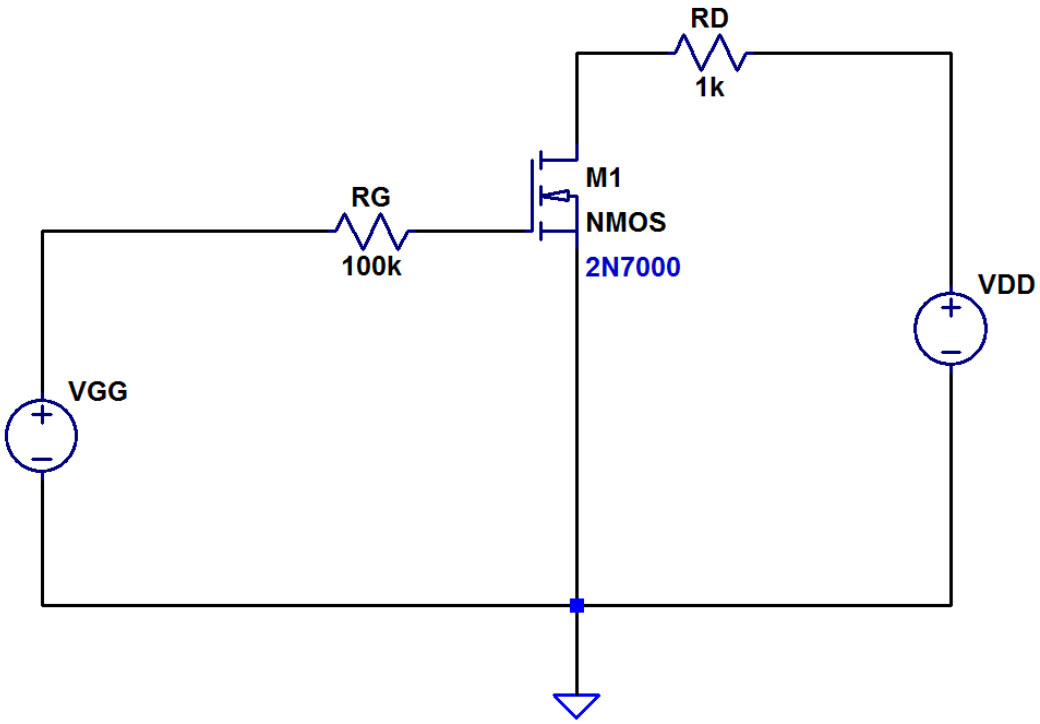


Figure 3 (a) $V_G = 0$, no biasing between Gate & Source, 3 (b) $V_G > V_T$, forward bias to Gate w.r.t to Source. The -ve charge is pulled towards Gate. Thus, at a particular point number of electrons > number of holes within the depletion region. So, there is a creation of a region where n-type conductivity opposed to p-type is formed. This is called the inversion region. Hence n-channel is formed through which electrons can flow. 3 (c) $V_D = V_{DSat} = V_{GS} - V_T$, transition from ohmic region to saturation region, 3 (d) $V_D > V_{DSat}$, pinch-off point moves towards the Source thus reducing the channel length. Under these conditions, the area between the pinch-off point and the drain is fully depleted with no inversion layer. Since this region has no positive free carriers, there is no possibility for electron-hole recombination if an electron enters the region from the electron-rich source and, if there is an electric field across the depletion zone, the electron can freely transit to the drain. The current through the device becomes controlled solely by the gate voltage under drain saturation conditions.

CIRCUIT DIAGRAM:



BREIF PROCEDURE:

OUTPUT/DRAIN CHARACTERISTICS:

1. Connect the circuit as per given diagram properly.
2. Keep V_{GS} constant at some value say 1.1 V by varying V_{GG}
3. Vary V_{DS} in step of 1V up to 10 volts and measure the drain current I_D. Tabulate all the readings.
4. Repeat the above procedure for V_{GS} as 1.2V, 1.3V, 1.4V, 1.5V etc

TRANSFER CHARACTERISTICS

1. Connect the circuit as per given diagram properly.
2. Set the voltage V_{DS} constant at 10 V.
3. Vary V_{GS} by varying V_{GG} in the step of 0.1 up to 1.55V and note down value of drain currentI_D. Tabulate all the readings.
4. Plot the output characteristics V_{DS} vs I_D and transfer characteristics V_{GS} vs I_D.
5. Calculate V_T, gm, rd or ro from the graphs and verify it from the data sheet